

very short timeline, it is better to go for a module because you have something almost there as a product."

Another deciding factor for modules versus chipsets depends on the kind of target volume a designer has in mind. Singh says, "Let us suppose you have a target volume in 1000s. The expenses to be incurred would include cost of designing a radio circuitry and certification costs."

There are certain frequency bands restricted for transmission by government, which are different for different countries. Ensuring the fact that it is working perfectly across various geographies, and making sure the device's harmonics do not interfere with restricted frequency bands, adds a lot of upfront cost to it, notes Singh. He adds, "It does not make sense to bear that cost for a volume in 1000s. But, if you are going to do a product to be manufactured in millions, then it makes sense to actually design everything from scratch."

For developers, it is going to be ease-of-interfacing and design-integration to the main board. Whether chipset provides that ease or module does, that would eventually be the deciding factor. However, Anand says, "It is expected that modules would be easier to interface and integrate, rather than chipsets, and they would also involve additional design care and complexities." He adds, "An integrated chipset will be a lower-cost solution, but with design and certification challenges. I think, modules would win over chipsets in the long run."

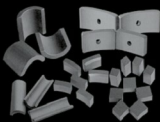
Modules will cost lesser in future

The better the integration, the more cost-effective the designer is going to be at a system level, notes Ganapathy. He says, "The chipset cost might actually increase as more and more features are integrated, because earlier you might have had to use a separate microcontroller and RF device. Now, you have one chip performing both functions, including integration of front-end modules." He adds, "So, as you integrate more, the system-on-chip may actually go up. But, when you look at the overall system cost, the price comes down. As volumes pick up, we could expect the prices of modules and chipsets to go down."

"Prices are decreasing as the volumes are increasing," comments Anand on the pricing of modules. However, until recently, prices seem to have stagnated and, perhaps, are waiting to change in some direction depending upon the market demand. He adds, "They are set to go down in the near future." However, there is one factor that may increase the price of modules and chipsets—functionalities and features. Anand adds, "If manufacturers could add more of those, they would be demanding premium and, hence, a bit higher price than basic counterparts." ●



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